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Liquid supply apparatus and substrate treatment apparatus including the same

Abstract

A liquid supply apparatus is provided. The liquid supply apparatus includes: a base unit having an operating space, an inlet port, through which a treatment liquid is introduced, and an outlet port, through which the treatment liquid is discharged, formed therein; bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space; an operating unit supplying operating fluid into or discharging the operating fluid from the operating space; and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, wherein the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space to replace the treatment liquid.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Korean Patent Application No. 10-2022-0159175 filed on Nov. 24, 2022 in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which in its entirety are herein incorporated by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to a liquid supply apparatus and a substrate treatment apparatus including the same.

2. Description of the Related Art

(3) A photo process for manufacturing semiconductor devices may involve forming a photoresist

layer on a wafer, followed by exposure and development. During this photo process, a coating apparatus may be used to form the photoresist layer on the wafer or to apply a developing solution for shaping the photoresist layer.

(4) The coating apparatus may include a pump for supplying treatment liquids, such as photoresist and developing solutions. In other words, the coating apparatus can apply various treatment liquids onto the wafer through fluid pumping.

(5) Meanwhile, the pump is driven by a motor, and when the motor starts, there is a temporary increase in load. To avoid frequent stopping and restarting of the motor within a short period of time, the motor continues to operate. Thus, to intermittently supply treatment liquid at each substrate treatment position while keeping the pump running, a treatment liquid circulation structure is necessary. The treatment liquid circulation structure prevents the continuous ejection of treatment liquid onto the substrate and instead allows for the recovery of the treatment liquid from the nozzles back to the storage tank. As a result, lines need to be connected to the pump, and additional components are required along those lines. However, this may introduce complexity to the overall treatment liquid circulation structure. Furthermore, during the replacement of the existing treatment liquid with a new treatment liquid, there is a potential risk of mixing the new and existing treatment liquids if there is residual treatment liquid inside the pump. This is particularly problematic when the new and existing treatment liquids differ in type or viscosity. Therefore, improvements are needed to address such and other issues.

SUMMARY

(6) Aspects of the present disclosure provide a liquid supply apparatus capable of supplying treatment liquid without using a motor and releasing any residual treatment liquid when replacing it with a new treatment liquid, and a substrate treatment apparatus including the liquid supply apparatus.

(7) However, aspects of the present disclosure are not restricted to those set forth herein. The above and other aspects of the present disclosure will become more apparent to one of ordinary skill in the art to which the present disclosure pertains by referencing the detailed description of the present disclosure given below.

(8) According to an aspect of the present disclosure, a liquid supply apparatus includes: a base unit having an operating space, an inlet port, through which a treatment liquid is introduced, and an outlet port, through which the treatment liquid is discharged, formed therein; bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space; an operating unit supplying operating fluid into or discharging the operating fluid from the operating space; and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, wherein the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space to replace the treatment liquid.

(9) According to another aspect of the present disclosure, a substrate treatment apparatus includes: a substrate support unit supporting a substrate; a cup portion surrounding the substrate support unit; a nozzle unit connected to a treatment liquid supply line and ejecting a treatment liquid onto the substrate; and a liquid supply apparatus provided in the treatment liquid supply line and supplying the treatment liquid to the nozzle unit, wherein the liquid supply apparatus includes a base unit having an operating space, an inlet port, through which the treatment liquid is introduced, and an outlet port, through which the treatment liquid is discharged, formed therein, bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a

storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space, an operating unit supplying operating fluid into or discharging the operating fluid from the operating space, and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space to replace the treatment liquid.

(10) According to another aspect of the present disclosure, a liquid supply apparatus has a treatment liquid supply line connected thereto and supplies the treatment liquid to a nozzle unit, which ejects the treatment liquid onto a substrate. The liquid supply apparatus includes: a base unit having an operating space, an inlet port, which is formed through an upper part of the base unit and through which treatment liquid is introduced, and an outlet port, which is formed through the upper part of the base unit adjacent to the inlet port and through which the treatment liquid is discharged, formed therein; bellows having the operating space formed on its outside within the base unit, including having an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space; an operating unit supplying operating fluid, which includes an inert gas or air, into or discharging the operating fluid from the operating space; and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, wherein the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space in a state where the contraction and expansion of the bellows have stopped with the bellows in its minimally contracted state.

(11) It should be noted that the effects of the present disclosure are not limited to those described above, and other effects of the present disclosure will be apparent from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects and features of the present disclosure will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in which:
- (2) FIG. 1 is a schematic diagram of a substrate treatment apparatus according to some embodiments of the present disclosure;
- (3) FIG. 2 is a schematic diagram of a liquid supply apparatus according to a first embodiment of the present disclosure;
- (4) FIG. 3 is a schematic diagram illustrating the contracted state of bellows in the liquid supply apparatus according to the first embodiment of the present disclosure;
- (5) FIG. 4 is a schematic diagram illustrating the expanded state of the bellows in the liquid supply apparatus according to the first embodiment of the present disclosure;
- (6) FIG. 5 is a schematic diagram illustrating a state in which the supply of a treatment liquid is stopped and the remaining treatment liquid is being discharged from the liquid supply device according to the first embodiment of the present disclosure;
- (7) FIG. 6 is a flowchart illustrating a cleaning process of the liquid supply device according to the first embodiment of the present disclosure;
- (8) FIG. 7 is a schematic diagram illustrating a state in which the remaining treatment liquid is discharged from the liquid supply device according to the first embodiment of the present disclosure;

- (9) FIG. **8** is a schematic diagram illustrating a state in which an inert gas is supplied inside the storage space of the liquid supply device according to the first embodiment of the present disclosure;
- (10) FIG. **9** is a schematic diagram illustrating a state in which a cleaning fluid is supplied inside the storage space of the liquid supply device according to the first embodiment of the present disclosure;
- (11) FIG. **10** is a schematic diagram illustrating a state in which a new processing liquid is supplied inside the storage space of the liquid supply device according to the first embodiment of the present disclosure;
- (12) FIG. **11** is a schematic diagram of a liquid supply device according to a second embodiment of the present disclosure; and
- (13) FIG. **12** is a schematic diagram illustrating a state in which an operating fluid is bypassed in the liquid supply device according to the second embodiment of the present disclosure.

DETAILED DESCRIPTION

- (14) Preferred embodiments of the present disclosure will hereinafter be described in detail with reference to the accompanying drawings. The advantages and features of the invention of the present disclosure, and methods for achieving them will become apparent from the detailed embodiments provided in conjunction with the accompanying drawings. However, it should be understood that the invention is not limited to the embodiments disclosed herein, but can be implemented in various other forms. The disclosed embodiments are provided to fully disclose the invention to those skilled in the art with ordinary knowledge in the field and are intended to define the scope of the invention solely by the claims. Throughout the specification, the same reference numerals refer to the same components.
- (15) The terminology used in this specification is for the purpose of describing the embodiments and should not be construed as limiting the invention. In this specification, the singular forms include the plural forms unless otherwise specified in the wording. The terms “comprises” and “comprising” used in the specification do not exclude the presence or addition of one or more other components, steps, operations, and/or elements as mentioned.
- (16) FIG. **1** is a schematic diagram of a substrate treatment apparatus according to some embodiments of the present disclosure.
- (17) Referring to FIG. **1**, a substrate treatment apparatus **10** may include cup portions **11**, substrate support units **12**, and nozzle units **14**. A treatment liquid may be supplied onto a substrate continuously or intermittently. For example, the treatment liquid may be a developing solution or a cleaning solution, but the present disclosure is not limited thereto.
- (18) The cup portions **11**, which recover the treatment liquid, may be provided to surround the substrate support units **12**. The cup portions **11** may be provided in the form of cylinders with open tops. Exhaust lines **15** for discharging the treatment liquid to the outside may be formed in the cup portions **11**.
- (19) The substrate support units **12** may support and rotate substrates within the cup portions **11**. Each of the substrate support units **12** may include a support plate **12P** and a driving member **12M**. The top surface of the support plate **12P** may have pins (not illustrated) for supporting a substrate. The support plate **12P** is rotatable by the driving member **12M**. The driving member **12M** may include a drive shaft and a driver. The drive shaft may be coupled to the bottom surface of the support plate **12P**. The driver may provide a rotational force to the drive shaft and may be a motor.
- (20) The nozzle units **14** may eject the treatment liquid onto the substrates within the cup portions **11**. Each of the nozzle units **14** may include one or more nozzles. If multiple nozzles are provided in each of the nozzle units **14**, treatment liquid supply lines **13** connected to a storage tank **13T** may be individually connected, allowing the multiple nozzles to independently eject the treatment liquid.
- (21) Although not specifically illustrated, the nozzle units **14** may be positioned at a waiting

position in a recessed port and may be designed to be movable between a processing position and the waiting position. The processing position refers to the position where the nozzle units **14** are facing the substrates on the support plates **12P**. The waiting position refers to the position where the nozzle units **14** remain in standby within the recessed port.

(22) A filter (not illustrated), a flow meter (not illustrated), a heater (not illustrated), a bubble cutter **13B**, an air vent **13V**, and a liquid supply apparatus **100** may be provided in each of the treatment liquid supply lines **13**, but the present disclosure is not limited thereto.

(23) The bubble cutter **13B** may be provided downstream of the liquid supply apparatus **100** in the corresponding treatment liquid supply line **13**. For example, the bubble cutter **13B** can remove bubbles generated during the flow of the treatment liquid.

(24) The air vent **13V** may be provided downstream of the liquid supply apparatus **100** in the corresponding treatment liquid supply line **13** to discharge and remove dissolved gases (e.g., air, N.sub.2, or O.sub.2) from the treatment liquid supply line **13**. For example, the air vent **13V** may be provided upstream of the bubble cutter **13B**. The air vent **13V** may remove larger bubbles, while the bubble cutter **13B** may remove finer bubbles. However, the present disclosure is not limited to this example, and various modifications and examples are possible.

(25) In another example, the air vent **13V** may not be positioned in the corresponding treatment liquid supply line **13**, but may be provided as a vent port above a base unit **110**, near an inlet port **113** and an outlet port **115** of the liquid supply apparatus **100**.

(26) The liquid supply apparatus **100** will hereinafter be described.

(27) FIG. 2 is a schematic diagram of a liquid supply apparatus according to a first embodiment of the present disclosure.

(28) Referring to FIG. 2, a liquid supply apparatus **100** may pump a treatment liquid, which flows along a treatment liquid supply line **13**, such that the treatment liquid may be supplied in a direction toward a nozzle unit **14**.

(29) The liquid supply device **100** may be driven by the supply of an operating fluid. In other words, the liquid supply device **100** may be driven by the supply and discharge of an operating fluid such as an inert gas (e.g., nitrogen) or air, rather than being driven by a cylinder or motor.

(30) For example, a typical pump is driven by a motor, and even if the treatment liquid does not process a substrate, the treatment liquid is continuously discharged. In other words, once the motor starts, a load is generated, and the motor cannot stop and restart. If the motor continues to operate without stopping, the treatment liquid may flow in the direction from the storage tank **13T** to the nozzle unit **14** even when the treatment liquid is not processing a substrate. To address this, a circulation structure may be implemented to recover the treatment liquid is recovered from the nozzle unit **14** to the storage tank **13T**, or alternatively, the treatment liquid may be discarded, resulting in wastage of the treatment liquid.

(31) On the contrary, the liquid supply apparatus **100**, driven by the operating fluid, may not supply the treatment liquid continuously. Hence, there may not be a requirement to circulate the treatment liquid, such as cleaning liquid, simplifying the structure of the liquid supply apparatus **100**. There is a possibility of bubble formation during the flow of the treatment liquid along a circulation line. However, by eliminating the circulation line, the risk of bubble generation can be reduced.

(32) Specifically, the liquid supply apparatus **100** may include a base unit **110**, bellows **120**, an operating unit **130**, and a drain line **140**.

(33) The base unit **110** may form the appearance of the liquid supply apparatus **100**, and an operating space **111**, an inlet port **113**, and an outlet port **115** may be formed in the base unit **110**.

(34) The operating space **111** may be hollow, the bellows **120** may be provided in the operating space **111**, and the operating fluid may be allowed to flow in and out of the bellows **120** for the operation of the bellows **120**.

(35) The inlet port **113** may receive the treatment liquid. For example, the inlet port **113** may penetrate an upper part of the base unit **110** and communicate with the interior of the bellows **120**.

In other words, the inlet port **113** may be connected to the treatment liquid supply line **13** and communicate with the bellows **120**, allowing the treatment liquid to flow through the treatment liquid supply line **13** to a storage space **121** of the bellows **120**. A lower part of the inlet port **113** may form the inlet of the bellows **120**.

(36) The outlet port **115** may discharge the treatment liquid. For example, the outlet port **115** may be adjacent to the inlet port **113** and penetrate the upper part of the base unit **110** to communicate with the interior of the bellows **120**. In other words, the outlet port **115** may be connected to the treatment liquid supply line **13** and communicate with the bellows **120**, allowing the treatment liquid to flow from the storage space **121** to the treatment liquid supply line **13** on a side of the nozzle unit **14**. The lower part of the outlet port **115** may form the outlet of the bellows **120**.

(37) The bellows **120** may expand or contract to adjust the volume of the storage space **121** and may supply the treatment liquid to the nozzle unit **14**. The inlet, the outlet, and the storage space **121** may be formed in the bellows **120** to allow the treatment liquid to flow in and out of the bellows **120**.

(38) The inlet port **113** may extend to the inlet of the bellows **120**, allowing the treatment liquid to enter the storage space **121** through the inlet of the bellows **120**. The inlet of the bellows **120** may be positioned adjacent to the outlet of the bellows **120**, at the top of the bellows **120**.

(39) The outlet portion **115** may extend to the outlet of the bellows **120** and may allow the treatment liquid to be discharged. The outlet of the bellows **120** may be provided at the top of the bellows **120** to ensure that air is discharged along with the treatment liquid without being accumulated in the storage space **121**.

(40) The storage space **121** may communicate with the inlet and the outlet of the bellows **120** such that the treatment liquid may flow in and out of the storage space **121**. The storage space **121** may be separated from the operating space **111**. For example, the bellows **120** may be provided inside the base unit **110**, and the operating space **111** may be formed on the outside of the bellows **120** such that the bellows **120** may expand or contract depending on the discharge and supply of the operating fluid. In other words, the bellows **120** may be provided inside the base unit **110** to be able to expand or contract in accordance with a variation in the inner pressure of the operating space **111**.

(41) The operating unit **130** may supply the operating fluid into or discharge the operating fluid out of the operating space **111**.

(42) The operating unit **130** may supply the operating fluid into the operating space **111** to allow the bellows **120** to contract. The operating unit **130** may discharge the operating fluid from the operating space **111** to allow the bellows **120** to expand. The treatment liquid may be discharged from the storage space **121** and supplied to the nozzle unit **14** through the supply and discharge of the operating fluid by the operating unit **130**.

(43) For example, the actuator **130** may include an operating fluid supply line **131** and an operating fluid discharge line **135**. The operating fluid supply line **131** and the operating fluid discharge line **135** may be installed to penetrate the base unit **110**, from below the operating space **111**, to establish communication with the operating space **111**, but the present disclosure is not limited thereto.

(44) The operating fluid supply line **131** may supply the operating fluid into the operating space **111** to allow the bellows **120** to contract. In other words, the operating fluid supply line **131** can increase the internal pressure of the operating space **111** to pressurize the bellows **120** in an upward direction. For example, the operating fluid supply line **131** may be equipped with a nitrogen supply device or a compressor that supplies compressed air, but the present disclosure is not limited thereto.

(45) The operating fluid discharge line **135** may discharge the operating fluid to allow the bellows **120** to expand. In other words, the operating fluid discharge line **135** can lower the internal pressure of the operating space **111**, causing the bottom surface of the bellows **120** to be pulled in a

downward direction. For example, the operating fluid discharge line **135** may be equipped with a valve for venting or an ejector.

(46) The drain line **140** communicates with the storage space **121** from below the base unit **110** and may discharge the treatment liquid. Although not specifically illustrated, the drain line **140** may be equipped with a valve for flow control.

(47) The drain line **140** may discharge the residual treatment liquid in the storage space **121** to facilitate the replacement of the treatment liquid. For example, the drain line **140** may discharge the residual treatment liquid with the bellows **120** in its minimally contracted state, minimizing or preventing the mixing of a newly supplied treatment liquid with the existing treatment liquid.

(48) The supply of the treatment liquid and the discharge of the residual treatment liquid will hereinafter be described.

(49) FIGS. **3** through **5** are diagrams illustrating the operations of the bellows and the drain line of the liquid supply apparatus of FIG. **2**.

(50) Referring to FIGS. **3** and **4**, the liquid supply device **100** allows the bellows **120** to expand or contract, thereby changing the volume of the storage space **121**, and as a result, the treatment liquid may be supplied to the nozzle unit **14**.

(51) As illustrated in FIG. **3**, the operating fluid supply line **131** may supply the operating fluid into the operating space **111**, causing the bellows **120** to contract. The contraction of the bellows **120** leads to an increase in the internal pressure within the operating space **111**, pressurizing the bottom surface of the bellows **120** upward. As a result, the volume of the storage space **121** within the bellows **120** may decrease. Consequently, the treatment liquid within the storage space **121** may be supplied to the nozzle unit **14** through the outlet port **115**.

(52) As illustrated in FIG. **4**, the bellows **120** may expand, allowing the storage space **121** to be filled with the treatment liquid. For this purpose, the operating fluid may be discharged from the operating space **111** through the operating fluid discharge line **135**. The expansion of the bellows **120** leads to an increase in the volume of the storage space **121**. With the expansion of the storage space **121**, the treatment liquid may be introduced into the storage space **121** through the inlet port **113**.

(53) Thus, the treatment liquid can be supplied to the nozzle unit **14** through the expansion and contraction of the bellows **120** driven by the operating unit **130**.

(54) Furthermore, when the treatment liquid is provided as a cleaning solution (or a developing solution), the treatment liquid may not be continuously ejected, but the flow of the treatment liquid may be stopped. For example, during standby at the recessed port, the treatment liquid can be prevented from being continuously discharged, thereby avoiding wastage of the treatment liquid.

(55) In other words, the supply of the treatment liquid to the nozzle unit **14** may be halted. For this purpose, the flow of the operating fluid may be stopped. For example, the operating fluid supply line **131** and the operating fluid discharge line **135** may both be closed to maintain a constant internal pressure in the operating space **111**.

(56) The liquid supply device **100** allows for the replacement of the treatment liquid. Since the properties or viscosity of the treatment liquid may vary, it is desirable to supply a new treatment liquid after emptying the remaining treatment liquid.

(57) Referring to FIG. **5**, the remaining treatment liquid in the storage space **121** may be discharged before supplying a new treatment liquid.

(58) For this purpose, in a state where the expansion or contraction of the bellows **120** is halted, in other words, in a state where the internal pressure of the operating space **111** remains unchanged, the flow path of the drain line **140** may be opened to allow the remaining treatment liquid to be discharged.

(59) The state where the internal pressure of the operating space **111** remains unchanged corresponds to the condition where the treatment liquid is not supplied and the discharge of the treatment liquid to the nozzle unit **14** is stopped. To this end, the flow paths of both the operating

fluid supply line **131** and the operating fluid discharge line **135** may be closed, thereby halting the flow of the operating fluid.

(60) In this manner, the drain line **140** can discharge the remaining treatment liquid in the storage space **121** with the expansion or contraction of the bellows **120** halted. The drain line **140**, which communicates with the storage space **121** at the bottom of the bellows **120**, can discharge the treatment liquid, including the liquid on the inner bottom surface of the storage space **121**, minimizing the amount of remaining treatment liquid.

(61) During the replacement of the treatment liquid, the storage space **121** of the bellows **120** may be rinsed, and this will hereinafter be described with reference to FIGS. **6** through **10**.

(62) FIG. **6** is a flowchart illustrating a cleaning operation for the liquid supply apparatus of FIG. **2**. FIGS. **7** through **10** are diagrams illustrating how to rinse the liquid supply apparatus of FIG. **2** and supply a new treatment liquid.

(63) Referring to FIGS. **6** through **10**, the following processes may be performed to replace the existing treatment liquid within the storage space **121** with a new treatment liquid.

(64) The storage space **121** may be supplied with an inert gas and a clean fluid, such as a cleaning fluid, through the inlet port **113**. The clean fluid may be discharged through the drain line **140**, instead of the outlet port **115**, which is intended for supplying the treatment liquid to the nozzle unit **14**. However, the present disclosure is not limited to this. Alternatively, the clean liquid may be discharged through the outlet port **115**, and the treatment liquid supply line **13**, located downstream of the outlet port **115**, may be rinsed.

(65) Referring to FIGS. **6** and **7**, as already mentioned with reference to FIG. **5**, the flow path of the drain line **140** may be opened to discharge the remaining treatment liquid on the inner bottom surface of the storage space **121** (**S110**).

(66) To minimize the waste of the treatment liquid, the remaining treatment liquid may be discharged when the bellows **120** is in its minimally contracted state. However, the present disclosure is not limited to this, and various modifications are possible. Alternatively, the remaining treatment liquid may be discharged when the bellows **120** is maximally expanded. Furthermore, the bellows **120** may be maximally expanded after discharging the remaining treatment liquid in its minimally contracted state.

(67) Thereafter, referring to FIGS. **6** and **8**, to minimize the use of a cleaning fluid, an inert gas may be supplied to the storage space **121** before supplying the cleaning fluid (**S120**). For example, nitrogen may be supplied through the inlet port **113** and may be released through the drain line **140** to purge the storage space **121**.

(68) Thereafter, referring to FIGS. **6** and **9**, the storage space **121** may be rinsed by supplying a cleaning fluid into the storage space **121** (**S130**). The cleaning fluid may be isopropyl alcohol (IPA). The cleaning fluid may differ from a cleaning solution for the treatment liquid, and the cleaning solution may be a thinner. However, the present disclosure is not limited to this.

(69) For a cleaning operation using the cleaning fluid, the cleaning fluid may be supplied through the inlet port **113** and discharged through the drain line **140** to clean the storage space **121**.

(70) Thereafter, referring to FIG. **6**, to ensure that no cleaning fluid remains in the storage space **121**, the storage space **121** may be purged again (**S140**).

(71) Thereafter, referring to FIGS. **6** and **10**, a new treatment liquid may be supplied to the storage space **121** (**S150**). When supplying the new treatment liquid to the storage space **121** through the inlet port **113**, the flow path of the drain line **140** may be temporarily opened at an early stage to purge nitrogen from the storage space **121**. After all the nitrogen has been purged from the storage space **121**, the flow path of the drain line **140** may be closed to allow the new treatment liquid to be supplied to the nozzle unit **14**. However, the present disclosure is not limited to this.

(72) A liquid supply apparatus according to another embodiment of the present disclosure will hereinafter be described with reference to FIGS. **11** and **12**, and redundant descriptions of duplicate configurations with the same functions are omitted.

(73) FIG. 11 is a schematic diagram of a liquid supply apparatus according to a second embodiment of the present disclosure. FIG. 12 is a schematic diagram illustrating the bypass of an operating fluid in the liquid supply apparatus of FIG. 11. The embodiment of FIGS. 11 and 12 will hereinafter be described, highlighting the differences with the embodiments of FIGS. 2 through 10.

(74) Referring to FIGS. 11 and 12, a liquid supply apparatus 100, like its counterpart of FIGS. 2 through 10, may include a base unit 110, bellows 120, an operating unit 130, and a drain line 140.

(75) In the embodiment of FIGS. 11 and 12, unlike in the embodiment of FIGS. 2 through 10, the operating unit 130 further includes a bypass line 133. That is, the operating unit 130 may include an operating fluid supply line 131, the bypass line 133, and an operating fluid discharge line 135.

(76) The bypass line 133 may diverge from the operating fluid supply line 131 and may enable a treatment liquid to bypass. Thus, an operating fluid can be redirected through the bypass line 133 and can thus pass through the operating fluid supply line 131 without increasing the internal pressure within an operating space 111, as the operating fluid. The purpose of bypassing the operating fluid via the bypass line 133 is to prevent the contraction of the bellows 120.

(77) Referring to FIG. 12, even when the operating fluid passes through the operating fluid supply line 131, the bypass line 133 may be opened to prevent the treatment liquid from being ejected from a nozzle unit 13 in a recessed port, and as a result, the operating fluid may be discharged through the bypass line 133.

(78) Consequently, the operating fluid is not supplied to the operating space 111 so that the contraction of the bellows 120 may be prevented. Furthermore, the flow path of the operating fluid discharge line 135 may be closed. Therefore, the expansion of the bellows 120 can also be prevented because the internal pressure within the operating space 111 does not change.

(79) In the first embodiment, the operating unit 130 closes both the flow paths of the operating fluid supply line 131 and the operating fluid discharge line 135 to prevent any change in the volume of a storage space 121. On the contrary, in the second embodiment, the operating unit 130 allows the operating fluid to flow through the operating fluid supply line 133, but bypass the storage space 121 through the bypass line 133 without entering the storage space 121. In other words, various modifications are possible to achieve a state where the internal pressure within the operating space 111 remains unchanged.

(80) Additionally, although not specifically illustrated, the bypass line 133 may rejoin the operating fluid supply line 131 for circulation. Since the operating fluid, unlike the treatment liquid, does not require consideration for bubble reformation issues, the circulation line of the operating fluid does not need a configuration to prevent bubble reformation or remove bubbles, simplifying the structure of the liquid supply apparatus 100.

(81) Embodiments of the present disclosure have been described with reference to the accompanying drawings. However, it should be understood that the present disclosure is not limited to these embodiments and can be implemented in various other forms. Those skilled in the art to which the present disclosure pertains, with ordinary knowledge in the field, will understand that the technical concepts or essential features of the present disclosure can be implemented in different specific forms without altering their essence. Therefore, the embodiments described above should be considered exemplary and not restrictive in any way.

Claims

1. A liquid supply apparatus comprising: a base unit having an operating space, an inlet port, through which a treatment liquid is introduced, and an outlet port, through which the treatment liquid is discharged, formed therein; bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage

space; an operating unit supplying an operating fluid into or discharging the operating fluid from the operating space; and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, wherein the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space to replace the treatment liquid.

2. The liquid supply apparatus of claim 1, wherein the outlet of the bellows is provided at a top of the bellows to discharge air along with the treatment liquid without being accumulated in the storage space.

3. The liquid supply apparatus of claim 2, wherein the inlet of the bellows is provided at the top of the bellows, adjacent to the outlet of the bellows.

4. The liquid supply apparatus of claim 1, wherein the operating unit includes an operating fluid supply line, which supplies the operating fluid to allow the bellows to expand, and an operating fluid discharge line, which discharges the operating fluid to allow the bellows to contract.

5. The liquid supply apparatus of claim 4, wherein the operating unit further includes a bypass line, which diverges from the operating fluid supply line and bypasses the treatment liquid.

6. The liquid supply apparatus of claim 1, wherein the operating fluid includes an inert gas or air.

7. The liquid supply apparatus of claim 1, wherein the treatment liquid includes a cleaning fluid or a developing solution.

8. The liquid supply apparatus of claim 1, wherein the drain line discharges the treatment liquid from the storage space with the bellows in its minimally contracted state.

9. A substrate treatment apparatus comprising: a substrate support unit supporting a substrate; a cup portion surrounding the substrate support unit; a nozzle unit connected to a treatment liquid supply line and ejecting a treatment liquid onto the substrate; and a liquid supply apparatus provided in the treatment liquid supply line and supplying the treatment liquid to the nozzle unit, wherein the liquid supply apparatus includes a base unit having an operating space, an inlet port, through which the treatment liquid is introduced, and an outlet port, through which the treatment liquid is discharged, formed therein, bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space, an operating unit supplying an operating fluid into or discharging the operating fluid from the operating space, and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space to replace the treatment liquid.

10. The substrate treatment apparatus of claim 9, wherein the outlet of the bellows is provided at a top of the bellows.

11. The substrate treatment apparatus of claim 10, further comprising: a bubble cutter provided downstream of the liquid supply apparatus and removing bubbles.

12. The substrate treatment apparatus of claim 10, further comprising: an air vent provided downstream of the liquid supply apparatus, in the treatment liquid supply line, and discharging dissolved gases.

13. The substrate treatment apparatus of claim 10, wherein the inlet of the bellows is provided at the top of the bellows, adjacent to the outlet of the bellows.

14. The substrate treatment apparatus of claim 9, wherein the operating unit includes an operating fluid supply line, which supplies the operating fluid to allow the bellows to expand, and an operating fluid discharge line, which discharges the operating fluid to allow the bellows to contract.

15. The substrate treatment apparatus of claim 14, wherein the operating unit further includes a

bypass line, which diverges from the operating fluid supply line and bypasses the treatment liquid.

16. The substrate treatment apparatus of claim 9, wherein the operating fluid includes an inert gas or air.

17. The substrate treatment apparatus of claim 9, wherein the treatment liquid includes a cleaning fluid or a developing solution.

18. The substrate treatment apparatus of claim 9, wherein the drain line discharges the treatment liquid from the storage space with the bellows in its minimally contracted state.

19. A liquid supply apparatus having a treatment liquid supply line connected thereto and supplying the treatment liquid to a nozzle unit, which ejects the treatment liquid onto a substrate, the liquid supply apparatus comprising: a base unit having an operating space, an inlet port, which is formed through an upper part of the base unit and through which treatment liquid is introduced, and an outlet port, which is formed through the upper part of the base unit adjacent to the inlet port and through which the treatment liquid is discharged, formed therein; bellows having the operating space formed on its outside within the base unit, including an inlet and an outlet, which form ends of the inlet and outlet ports, respectively, of the base unit and communicate to create a storage space that the treatment liquid flows in and out of, and expanding or contracting to increase or reduce the volume of the storage space; an operating unit supplying an operating fluid, which includes an inert gas or air, into or discharging the operating fluid from the operating space; and a drain line communicating with the storage space, from below the base unit, and discharging the treatment liquid, wherein the operating unit supplies the operating fluid into the operating space to enable the bellows to contract, or discharges the operating fluid from the operating space to enable the bellows to expand, and the drain line discharges residual treatment liquid remaining in the storage space in a state where the contraction and expansion of the bellows have stopped with the bellows in its minimally contracted state.
